AI-POWERED CREDIT RISK
INTELLIGENCE PLATFORM

NeoStats AI Engineer Assignment — Home Credit Default Risk

LIGHTGBM SHAP GEMINI LLM POSTGRES DOCKER STREAMLIT

Real dataset (307,511 applicants) · end-to-end, dockerized, verified

WHY THIS MATTERS

Faster, Explainable Credit Decisions

The business problem

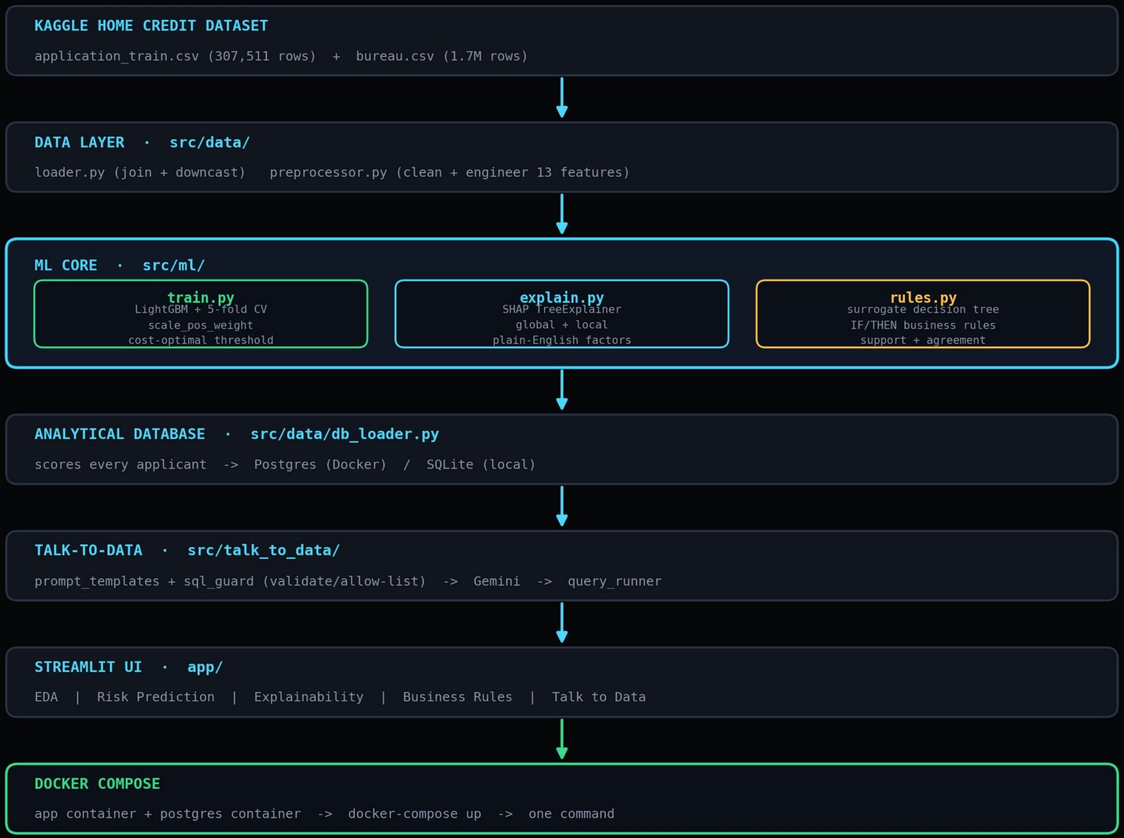
- Identify high-risk loan applicants early
- Automate risk scoring and decision support
- Provide explainable reasons for every classification
- Satisfy audit and regulatory requirements
- Let analysts explore data in plain English
- Bridge ML insights with real credit policy

What we delivered

- ✓ EDA with 5 business insights
- ✓ LightGBM model + evaluation metrics
- ✓ SHAP explainability, per-applicant
- ✓ Auto-derived business rules
- ✓ Gemini-powered talk-to-data chatbot
- ✓ Dockerized, one-command deployment

SYSTEM DESIGN

End-to-End Architecture



Every layer is independently testable — the Streamlit UI is a thin wrapper over src/, not where the logic lives.

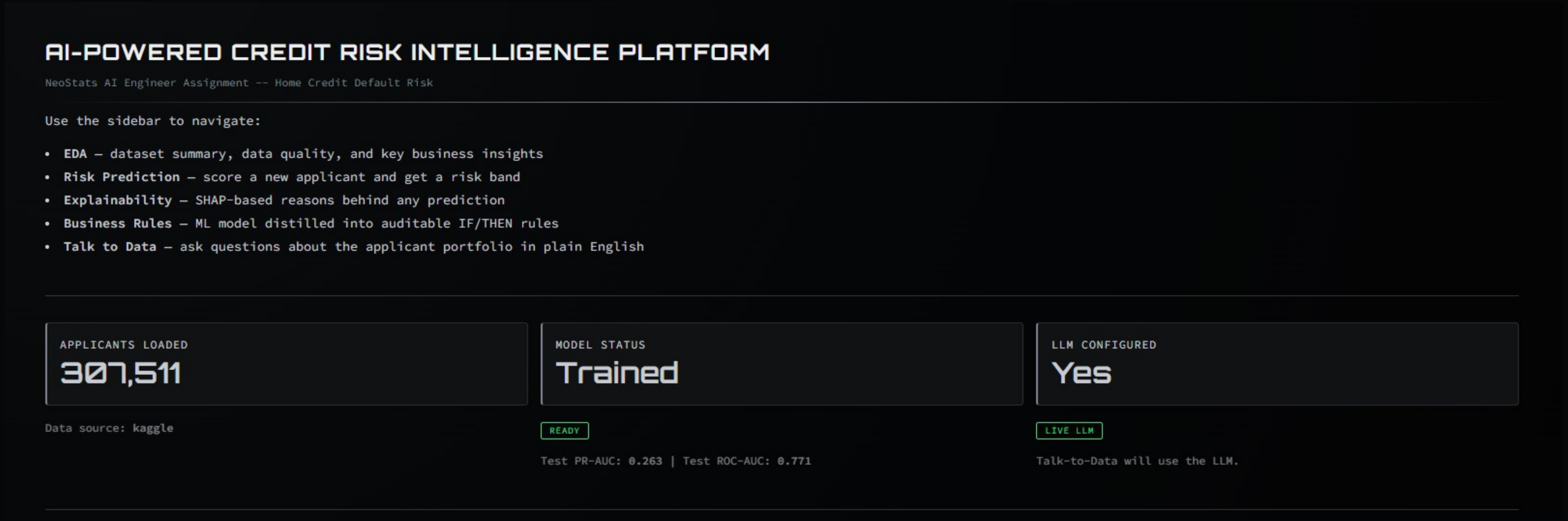
ASSIGNMENT MAPPING

Every Weighted Criterion, Covered

Evaluation Area	Weight	What We Built
ML solution design & model quality	30%	LightGBM + 5-fold CV, scale_pos_weight, cost-optimal threshold, PR-AUC-driven selection
LLM integration & NL-to-SQL	25%	Gemini-backed chatbot + 10 built-in fallback query patterns, zero-key operation
Prompt optimization & hallucination control	15%	sqlglot validation, table/column allow-list, mandatory LIMIT, response caching
EDA	15%	134-feature categorization, missing-value audit, 5 business insights with charts
Dockerization & engineering quality	10%	Multi-stage Dockerfile, docker-compose (app + Postgres), 17 passing unit tests
Documentation & architecture clarity	5%	Full README, Colab guide, data-source docs, inline design-rationale comments

LIVE PLATFORM

One Platform, Five Sections



Real numbers from the real dataset on load — 307,511 applicants, a trained model, and a live Gemini connection.

PART 1 · EDA

307K Applicants, 134 Features

Dataset summary

ROWS	COLUMNS	DEFAULT RATE	MISSING CELLS
307,511	134	8.07%	9,319,407

Feature categorization

Numeric features (118)

SK_ID_CURR, TARGET, CNT_CHILDREN, AMT_INCOME_TOTAL, AMT_CREDIT, AMT_ANNUITY, AMT_GOODS_PRICE, REGION_POPULATION_RELATIVE, DAYS_BIRTH, DAYS_EMPLOYED, DAYS_REGISTRATION, DAYS_ID_PUBLISH, OWN_CAR_AGE, FLAG_MOBIL, FLAG_EMP_PHONE, FLAG_WORK_PHONE, FLAG_CONT_MOBILE, FLAG_PHONE, FLAG_EMAIL, CNT_FAM_MEMBERS, REGION_RATING_CLIENT, REGION_RATING_CLIENT_W_CITY, HOUR_APPR_PROCESS_START, REG_REGION_NOT_LIVE_REGION, REG_REGION_NOT_WORK_REGION, LIVE_REGION_NOT_WORK_REGION, REG_CITY_NOT_LIVE_CITY, REG_CITY_NOT_WORK_CITY, LIVE_CITY_NOT_WORK_CITY, EXT_SOURCE_1, EXT_SOURCE_2, EXT_SOURCE_3, APARTMENTS_AVG, BASEMENTAREA_AVG, YEARS_BEGINEXPLUATATION_AVG, YEARS_BUILD_AVG, COMMONAREA_AVG, ELEVATORS_AVG, ENTRANCES_AVG, FLOORSMAX_AVG, FLOORSMIN_AVG, LANDAREA_AVG, LIVINGAPARTMENTS_AVG, LIVINGAREA_AVG, NONLIVINGAPARTMENTS_AVG, NONLIVINGAREA_AVG, APARTMENTS_MODE, BASEMENTAREA_MODE, YEARS_BEGINEXPLUATATION_MODE, YEARS_BUILD_MODE, COMMONAREA_MODE, ELEVATORS_MODE, ENTRANCES_MODE, FLOORSMAX_MODE, FLOORSMIN_MODE, LANDAREA_MODE, LIVINGAPARTMENTS_MODE, LIVINGAREA_MODE, NONLIVINGAPARTMENTS_MODE, NONLIVINGAREA_MODE, APARTMENTS_MEDI, BASEMENTAREA_MEDI, YEARS_BEGINEXPLUATATION_MEDI, YEARS_BUILD_MEDI, COMMONAREA_MEDI, ELEVATORS_MEDI, ENTRANCES_MEDI, FLOORSMAX_MEDI, FLOORSMIN_MEDI, LANDAREA_MEDI, LIVINGAPARTMENTS_MEDI, LIVINGAREA_MEDI, NONLIVINGAPARTMENTS_MEDI, NONLIVINGAREA_MEDI, TOTALAREA_MODE, OBS_30_CNT_SOCIAL_CIRCLE, DEF_30_CNT_SOCIAL_CIRCLE, OBS_60_CNT_SOCIAL_CIRCLE, DEF_60_CNT_SOCIAL_CIRCLE, DAYS_LAST_PHONE_CHANGE, FLAG_DOCUMENT_2, FLAG_DOCUMENT_3, FLAG_DOCUMENT_4, FLAG_DOCUMENT_5, FLAG_DOCUMENT_6, FLAG_DOCUMENT_7, FLAG_DOCUMENT_8, FLAG_DOCUMENT_9, FLAG_DOCUMENT_10, FLAG_DOCUMENT_11, FLAG_DOCUMENT_12, FLAG_DOCUMENT_13, FLAG_DOCUMENT_14, FLAG_DOCUMENT_15, FLAG_DOCUMENT_16, FLAG_DOCUMENT_17, FLAG_DOCUMENT_18, FLAG_DOCUMENT_19, FLAG_DOCUMENT_20, FLAG_DOCUMENT_21, AMT_REQ_CREDIT_BUREAU_HOUR, AMT_REQ_CREDIT_BUREAU_DAY, AMT_REQ_CREDIT_BUREAU_WEEK, AMT_REQ_CREDIT_BUREAU_MON, AMT_REQ_CREDIT_BUREAU_QRT, AMT_REQ_CREDIT_BUREAU_YEAR, AGE_YEARS, EMPLOYED_YEARS, EMPLOYED_TO_AGE_RATIO, CREDIT_TO_INCOME_RATIO, ANNUITY_TO_INCOME_RATIO, CREDIT_TO_GOODS_RATIO, INCOME_PER_FAMILY_MEMBER, ANNUITY_TO_CREDIT_RATIO, EXT_SOURCE_MEAN, EXT_SOURCE_MAX, EXT_SOURCE_MIN, EXT_SOURCE_MISSING_COUNT

Categorical features (16)

NAME_CONTRACT_TYPE, CODE_GENDER, FLAG_OWN_CAR, FLAG_OWN_REALTY, NAME_TYPE_SUITE, NAME_INCOME_TYPE, NAME_EDUCATION_TYPE, NAME_FAMILY_STATUS, NAME_HOUSING_TYPE, OCCUPATION_TYPE, WEEKDAY_APPR_PROCESS_START, ORGANIZATION_TYPE, FONDKAPREMONT_MODE, HOUSETYPE_MODE, WALLSMATERIAL_MODE, EMERGENCYSTATE_MODE

118 numeric + 16 categorical features, auto-categorized straight from the raw schema — no manual column typing.

Missingness Is Signal, Not Noise

No imputation

LightGBM splits on missing values natively — filling them would destroy real signal.

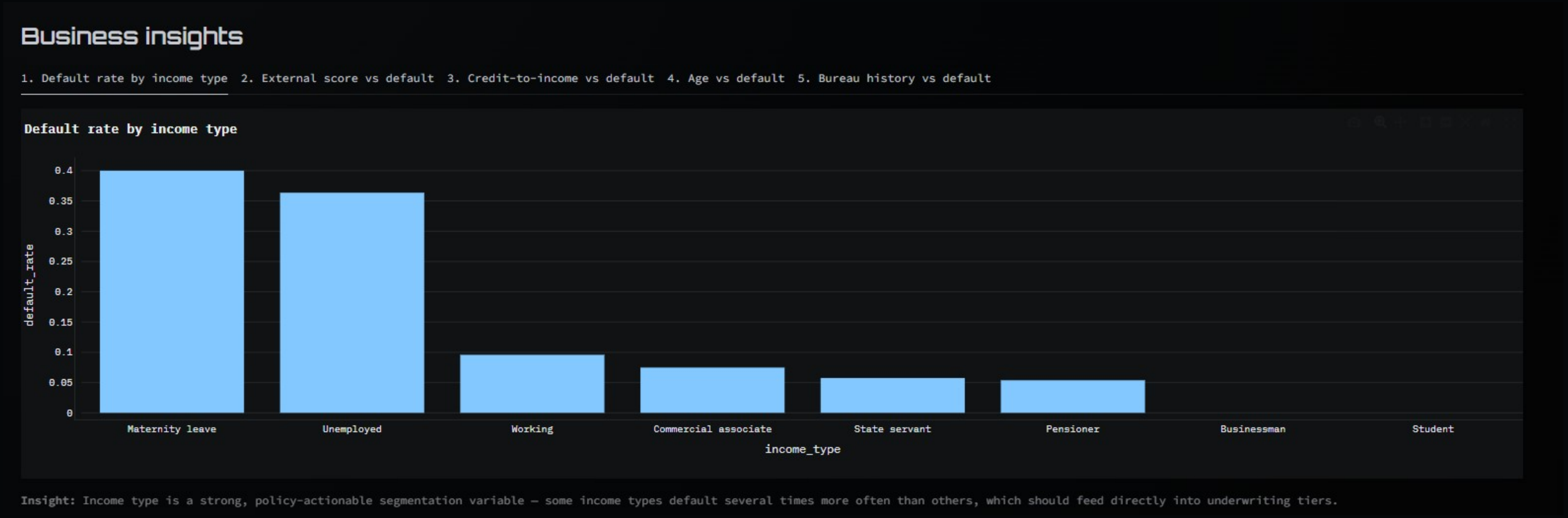
Housing metadata

Top missing columns describe apartment/building details, not core risk fields.

EXT_SOURCE_1 gap

~56% missing by design in the real dataset — the gap itself is mildly predictive.

Income Type Predicts Default Risk



Insight: Maternity leave and Unemployed applicants default 4–6x more often than Working or State servant applicants — a strong, policy-actionable underwriting signal.

4 more insights (external score, credit-to-income, age, bureau history) are live in the app's EDA page.

Cost-Optimal, Not Accuracy-Optimal

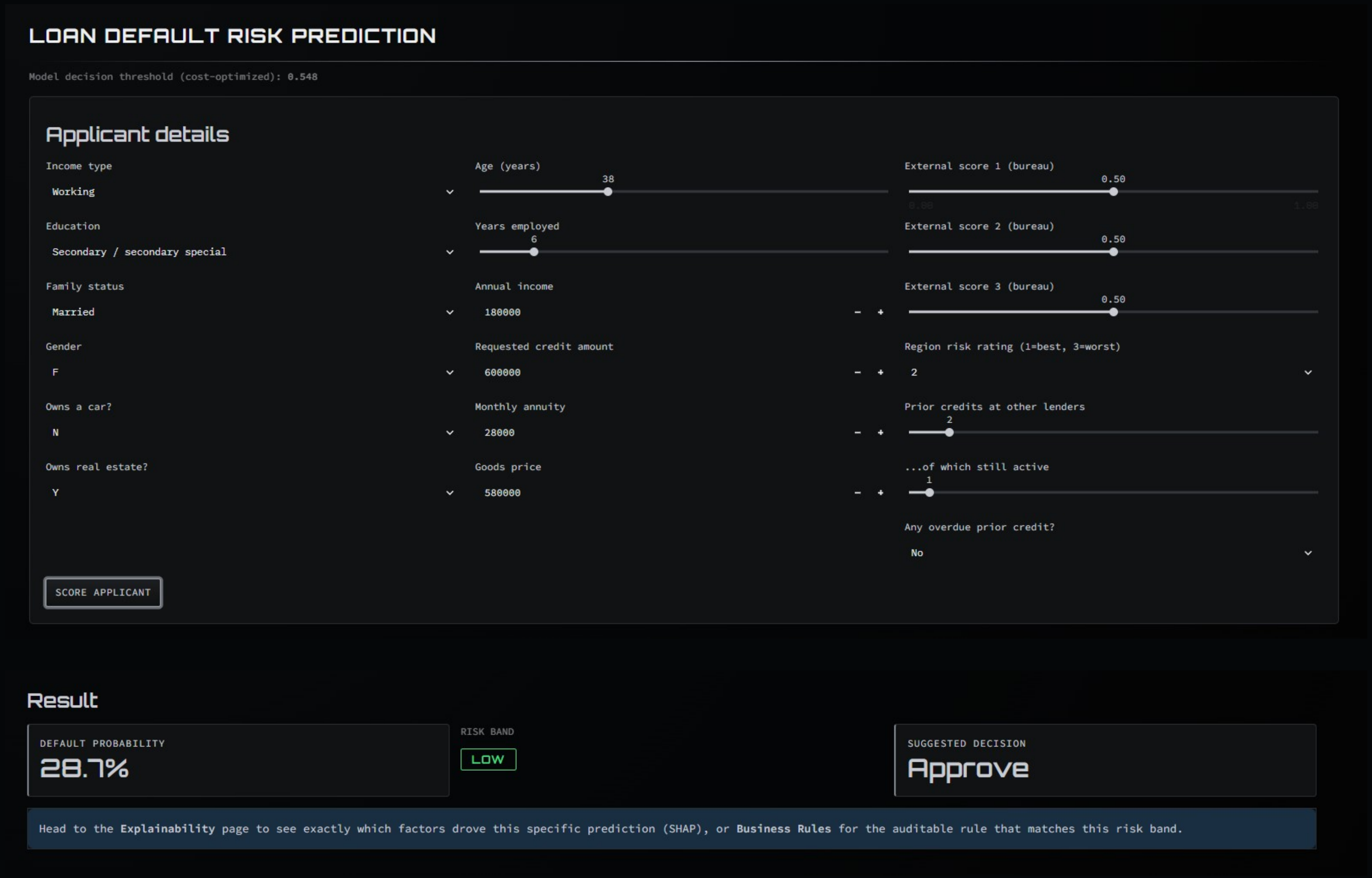
```
!python -m src.ml.evaluate

================================================================
CREDIT RISK MODEL -- EVALUATION REPORT
================================================================
Trained at (UTC):         2026-09-05T15:58:48.708381+00:00
Data source:              kaggle
Train/val rows:           246008
Held-out test rows:       61503
Base default rate:        8.07%
----------------------------------------------------------------
Out-of-fold PR-AUC:       0.2508
Out-of-fold ROC-AUC:      0.7649
Held-out test PR-AUC:     0.2626
Held-out test ROC-AUC:    0.7712
----------------------------------------------------------------
Decision threshold:       0.548  (cost-optimal, FN weighted heavier than FP)
  -> precision @ thresh: 0.200
  -> recall @ thresh:    0.578
----------------------------------------------------------------
Confusion matrix on held-out test set:
  True Positive (correctly flagged defaulters): 2994
  False Negative (missed defaulters):            1971
  False Positive (good applicants declined):     11809
  True Negative (correctly approved):            44729
================================================================
```

- LightGBM + stratified 5-fold CV, early stopping
- scale_pos_weight for imbalance — not SMOTE
- Model selection on PR-AUC, not raw accuracy
- Decision threshold picked by minimizing business cost (FN 8x worse than FP), not a default 0.5 cutoff

0.771	0.263	0.548	57.8%
TEST ROC-AUC	TEST PR-AUC	DECISION THRESHOLD	RECALL @ THRESHOLD

Score Any Applicant in Seconds



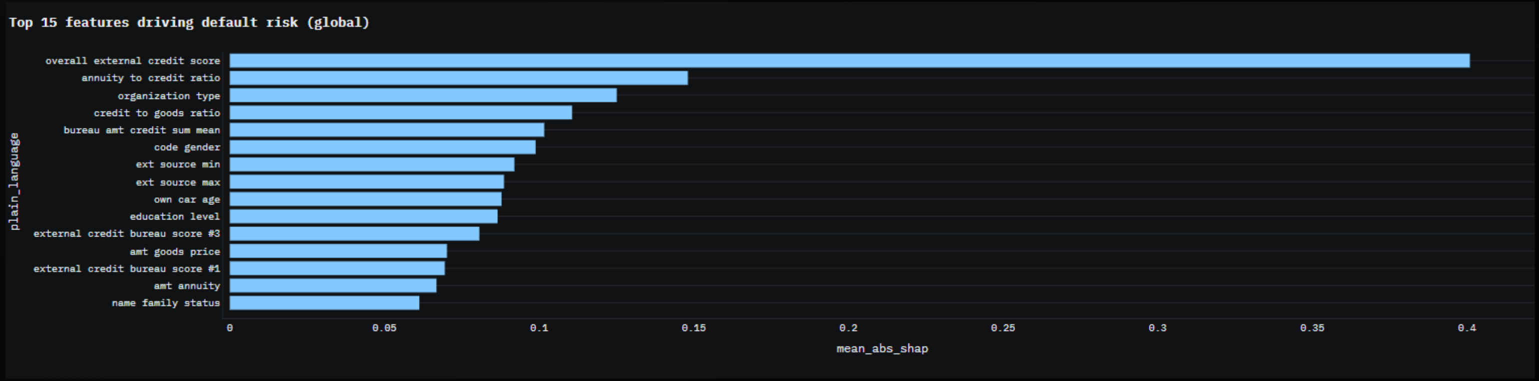
Every prediction pairs a probability with a plain risk band and a suggested decision — ready for a credit officer, not just a data scientist.

Every Prediction, Explained

EXPLAINABLE AI (SHAP)

Global: what drives risk portfolio-wide   Local: explain one applicant

Mean absolute SHAP value per feature across a sample of the portfolio — this is the model's own view of what matters, for governance and audit review.



	feature	mean_abs_shap	plain_language
0	EXT_SOURCE_MEAN	0.4009	overall external credit score
1	ANNUITY_TO_CREDIT_RATIO	0.1481	annuity to credit ratio
2	ORGANIZATION_TYPE	0.1251	organization type
3	CREDIT_TO_GOODS_RATIO	0.1106	credit to goods ratio
4	BUREAU_AMT_CREDIT_SUM_MEAN	0.1016	bureau amt credit sum mean
5	CODE_GENDER	0.0989	code gender
6	EXT_SOURCE_MIN	0.092	ext source min
7	EXT_SOURCE_MAX	0.0886	ext source max
8	OWN_CAR_AGE	0.0878	own car age
9	NAME_EDUCATION_TYPE	0.0866	education level

SHAP TreeExplainer, global + per-applicant — overall external credit score dominates, matching every published analysis of this dataset.

MODEL DISTILLATION

The Model, Distilled Into Policy

BUSINESS-READABLE DECISION RULES

These rules are **distilled from the trained ML model**, not hand-written — a shallow decision tree is fit to reproduce the model's own Low/Medium/High risk-band output, then converted into plain-English IF/THEN statements. Each rule reports:

- **Coverage (support):** how many applicants it applies to
- **Agreement:** what fraction of those applicants the rule correctly matches to the model's actual output (i.e. rule fidelity, not ground-truth accuracy)

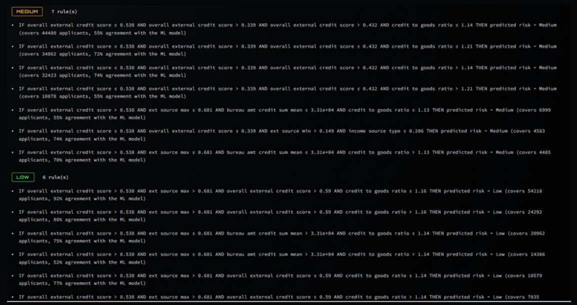
Use these to explain credit policy to non-technical stakeholders or auditors without exposing raw model internals.

RE-DERIVE RULES FROM CURRENT MODEL

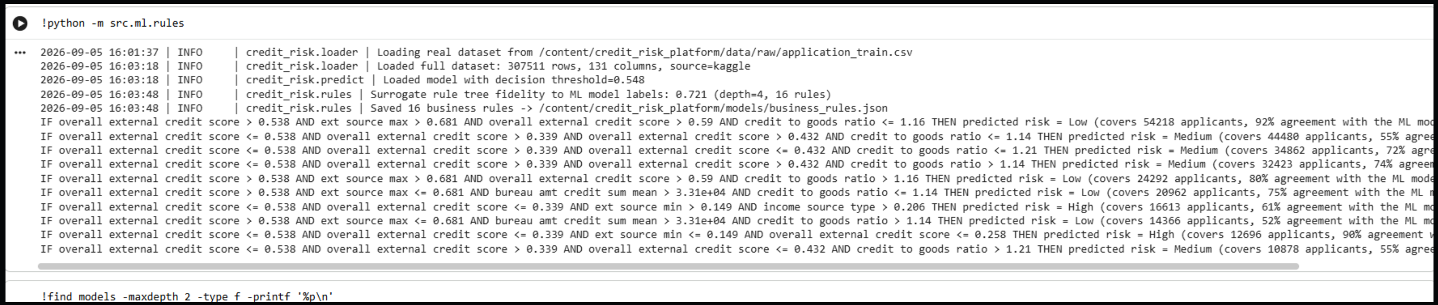
Filter by predicted band

HIGH 3 rule(s)

- IF overall external credit score ≤ 0.538 AND overall external credit score ≤ 0.339 AND ext source min > 0.149 AND income source type > 0.206 THEN predicted risk = High (covers 16613 applicants, 61% agreement with the ML model)
- IF overall external credit score ≤ 0.538 AND overall external credit score ≤ 0.339 AND ext source min ≤ 0.149 AND overall external credit score ≤ 0.258 THEN predicted risk = High (covers 12696 applicants, 90% agreement with the ML model)
- IF overall external credit score ≤ 0.538 AND overall external credit score ≤ 0.339 AND ext source min ≤ 0.149 AND overall external credit score > 0.258 THEN predicted risk = High (covers 8040 applicants, 68% agreement with the ML model)



Medium & Low bands



Live derivation run — 16 rules, 72% fidelity

PART 2 · TALK-TO-DATA

Ask the Portfolio a Question

What is the default rate by income type?

Here are the top results for your question (8 row(s) shown):

- Income type: Maternity leave, Default rate: 40.00%, Applicants: 5
- Income type: Unemployed, Default rate: 36.36%, Applicants: 22
- Income type: Working, Default rate: 9.59%, Applicants: 158,774
- Income type: Commercial associate, Default rate: 7.48%, Applicants: 71,617
- Income type: State servant, Default rate: 5.75%, Applicants: 21,703

› Show generated SQL

	name_income_type	default_rate	n
0	Maternity leave	0.4	5
1	Unemployed	0.3636	22
2	Working	0.0959	158774
3	Commercial associate	0.0748	71617
4	State servant	0.0575	21703
5	Pensioner	0.0539	55362
6	Businessman	0	10
7	Student	0	18

Which region has the highest risk?

Here are the top results for your question (3 row(s) shown):

- Region rating: 3.00, High-risk share: 23.98%, Applicants: 48,330.00
- Region rating: 2.00, High-risk share: 13.97%, Applicants: 226,984.00
- Region rating: 1.00, High-risk share: 5.36%, Applicants: 32,197.00

› Show generated SQL

	region_rating_client	high_risk_share	n
0	3	0.2398	48330
1	2	0.1397	226984
2	1	0.0536	32197

Natural language → validated SQL → plain-English answer. Every query passes table/column allow-listing before it ever touches the database.

SAFETY LAYER

Guardrails Before Generation

The enforcement layer is code, not a polite prompt request.

1. Parse with sqlglot
2. Table / column allow-list
3. Forbidden keyword scan
4. Mandatory LIMIT cap
5. Rolled-back transaction

Token optimization

- Condensed schema injection (~150 tokens) — not a full DDL dump or sample rows
- 6 shape-diverse few-shot examples, not exhaustive question coverage
- Response cache — repeated questions cost zero additional tokens
- Compact CSV results, never a full DataFrame dump, sent to the LLM

One Command, Two Paths

Docker (evaluator / production)

- docker-compose up — app + Postgres containers
- Multi-stage Dockerfile, non-root user, healthchecks
- Auto-trains on first boot, persists via named volumes
- Model artifact skips retraining on restart

Google Colab (training / verification)

- Mount Drive → extract real dataset → train on full 307K rows
- No Docker needed — SQLite backs the chatbot locally
- Streamlit tunneled out via cloudflared for a public URL

RESULTS

Results at a Glance

0.771	Test ROC-AUC
8.07%	Real base default rate
16	Business rules derived
10	Chatbot query patterns
17 / 17	Unit tests passing
307,511	Real applicants scored

HONEST SCOPE

Known Limitations & Future Work

Now

- Bureau features are simple aggregates — previous_application.csv and installments_payments.csv are not yet joined
- Surrogate rule tree is a deliberate simplification; it reports its own agreement rate rather than hiding disagreement
- Chatbot queries a single flattened table — no multi-table natural-language joins yet
- No model monitoring or drift detection

Next

- Wire in installment & POS-cash history for deeper features
- Formal fair-lending review — age and gender carry real signal and need compliance sign-off before real-world use
- Multi-table chatbot schema for richer natural-language joins
- Score-drift monitoring dashboard for production use

THANK YOU

AI-Powered Credit Risk Intelligence Platform — NeoStats AI Engineer Assignment

GITHUB

README

DOCKER-COMPOSE UP